

Sequence embeddings, not network topology,
predict interactions of uncharacterized proteins: a
leakage-aware benchmark with external validation
for reverse vaccinology in *Flavobacterium*

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Abstract

Background: Reverse vaccinology for bacterial fish pathogens increasingly combines protein prioritization with interaction networks, and graph neural networks (GNNs) are often proposed to predict missing protein–protein associations. For non-model pathogens, the proteins of greatest interest frequently have no recorded interactions, yet link-prediction models are usually evaluated on randomly held-out edges between well-connected proteins, with uniformly sampled

negatives. It is therefore unclear which inputs carry signal in the setting where such models are actually used.

Results: We built a homology-transferred STRING v12 network for the *Flavobacterium* sp. MO_0223_q7L1K proteome (3,257 proteins, 31,926 associations) and benchmarked three GNN encoders and a graph-free multilayer perceptron with Biomni agent-derived descriptors, ESM-2 protein language model embeddings and Node2Vec embeddings under three protocols, with 10 repeated splits each. On randomly held-out edges, parameter-free neighbourhood heuristics (ROC-AUC 0.969 ± 0.002) matched or exceeded every GNN (best 0.956 ± 0.005). When whole proteins were held out, Node2Vec fell to chance, and with uniform negatives the degree of the partner protein alone (0.860 ± 0.004) outperformed every trained model. With degree-matched negatives, all configurations combining ESM-2 with a graph-free encoder performed best and were statistically indistinguishable (ROC-AUC 0.767–0.777); no GNN improved on them. We validated this protocol externally on 728 proteins whose STRING links had been withheld by an upload limit (22,500 links): pooled ROC-AUC was 0.708, matching the internal estimate (0.715), and for per-protein partner lists plain ESM-2 nearest neighbours were the best pre-specified ranking (precision@5 0.135 versus 0.012 at random). Biomni priority tiers were not enriched for outer-membrane, TonB-dependent or type IX secretion families (High versus other tiers: odds ratio 0.61), and tier homophily in predicted links reflected model bias rather than the recovered interactions.

Conclusions: For proteins without known interactions, sequence embeddings provide the usable signal and graph structure provides none; evaluation protocol, not architecture, determined the apparent ranking of methods. We release a leakage-aware benchmark, an externally validated candidate list for 993 interaction-orphan proteins, and an audit of agent-derived vaccine priorities to support reverse vaccinology in *Flavobacterium*.

Keywords: Reverse vaccinology, Link prediction, Graph neural networks, Protein language models, Cold-start evaluation, *Flavobacterium*, Aquaculture

1 Background

Flavobacterium species cause some of the most damaging bacterial diseases of farmed freshwater fish, including columnaris disease (*F. columnare*) and bacterial cold-water disease (*F. psychrophilum*), and effective vaccines remain limited by incomplete knowledge of protective antigens [1–3]. Reverse vaccinology addresses this by ranking proteins from genome sequence for properties associated with protective immunity, such as surface exposure, secretion and conservation [4–6]. In *Flavobacterium*, the type IX secretion system (T9SS) is required for virulence and exports adhesins, peptidases and toxins bearing conserved C-terminal sorting domains [7–9], making secreted and outer-membrane proteins natural candidates.

Protein–protein association networks add biological context to such rankings, but they are incomplete and biased towards well-studied proteins [10, 11]. Link prediction, and in particular graph neural networks (GNNs) such as GCN, GAT and GraphSAGE

[12–14], has been proposed to infer missing associations, including for proteins without any known partner. Two well-documented evaluation problems make reported performance difficult to interpret for that use. First, randomly splitting edges leaves both proteins of almost every test pair in the training graph; performance on such pairs does not transfer to pairs involving unseen proteins [15], and models evaluated this way learn largely from sequence similarity and node degree [16]. Second, uniformly sampled negative pairs are easy to separate from positives on degree alone, so simple heuristics often match GNNs [17, 18]. Recent work further indicates that, once leakage is controlled, the signal in sequence-based interaction prediction comes mostly from protein language model (PLM) embeddings rather than from the downstream architecture [19].

Reverse vaccinology pipelines increasingly also rely on AI agents for protein annotation. Biomni is a general-purpose biomedical agent [20] that can produce per-protein physicochemical descriptors and priority scores, but the reliability of agent-derived priorities as antigen predictors has not been assessed against independent evidence.

Here we ask which inputs predict associations for proteins that have none recorded, using the proteome of a *Flavobacterium* isolate. We (i) benchmark Biomni descriptors, ESM-2 embeddings [21] and Node2Vec embeddings [22] with three GNN encoders, a graph-free encoder and non-learned baselines under protocols that do and do not hold out whole proteins and control for degree; (ii) validate the held-out-protein protocol externally on proteins whose associations were withheld from the development network; (iii) audit Biomni priority tiers against Pfam families of surface and secreted proteins chosen in advance; and (iv) release a ranked, annotated candidate list for proteins without interaction evidence.

2 Methods

2.1 Proteome, sequence groups and functional annotation

The study used the predicted proteome of *Flavobacterium* sp. MO_0223_q7L1K (3,257 proteins; RAST annotation, [AUTHOR INPUT: genome accession / BioProject and isolation source]). 320 proteins fall into 111 groups of identical sequence; members of a group receive identical features, so identical pairs were excluded from all candidate lists and external evaluation, and per-protein lists were collapsed to one representative per group. Protein domains were annotated with Pfam 38.2 [23] using pyhmmer [24] (HMMER3 [25]) with the curated gathering thresholds; domains were mapped to clans and to Gene Ontology terms with pfam2go. 2,300 proteins (71%) carried at least one domain.

2.2 Biomni descriptors and priority tiers

For every protein, Biomni [20] produced a signal-peptide score, the number of predicted transmembrane segments, cysteine content, GRAVY hydrophobicity, length, molecular weight, instability index, isoelectric point and a composite vaccine score, from which four priority tiers were assigned (Very Low < 0.3, Low 0.3–0.45, Medium 0.5–0.65, High ≥ 0.7) [AUTHOR INPUT: Biomni version, run date, language-model backbone,

prompts and the generated code, to be released as Supplementary File S1]. Because molecular weight and length are almost perfectly correlated ($r = 0.999$), molecular weight was dropped; the number of transmembrane segments and length were log-transformed, and all descriptors were standardised to zero mean and unit variance before modelling (node descriptors contain no edge information, so standardisation over all proteins does not leak test labels).

2.3 Protein language model embeddings

Each sequence was embedded with ESM-2 [21] (650M parameters, `esm2_t33.650M.UR50D`; 1,280 dimensions) by averaging final-layer residue representations, excluding special tokens. Sequences longer than 1,022 residues were split into consecutive chunks and all residues were averaged. The 35M-parameter model (480 dimensions) was used as a sensitivity analysis (Supplementary Table A1).

2.4 Interaction networks

Flavobacterium sp. MO_0223-q7L1K is not represented in STRING, so associations were transferred by homology from *F. columnare* ATCC 49512 (STRING taxon 1041826, STRING v12 [10]). Two networks were used (Table 1).

Development network. The STRING web interface accepts at most 2,000 sequences; the 2,000 longest proteins had been submitted, 1,760 were mapped by STRING, and the exported associations (combined score ≥ 0.4) formed a network of 16,315 edges in which 1,650 proteins had no edge, 1,257 of them because they were never submitted. All model development and selection used this network.

Final network. All 3,257 proteins were searched against the 2,642 STRING v12 proteins of taxon 1041826 with phmmer, keeping the best hit with a bit score at least as high as the weakest mapping STRING itself had accepted (51.2 bits). This mapping agreed with the STRING web mapping for 98.6% of the 1,760 proteins and mapped 745 of the 1,257 never-submitted proteins. Each STRING protein was assigned to its highest-scoring proteome match (lower-scoring paralogous matches remain without edges to avoid duplicating edge sets between near-identical proteins), and links with combined score ≥ 400 were retained, giving 31,926 edges and 993 proteins without edges. The development network’s edges are a subset of the final network (99.7% recovered), confirming that the original export also came from STRING v12.

2.5 Models and baselines

Three GNN encoders (GCN [12], GAT [13], GraphSAGE [14]) and a graph-free two-layer perceptron (MLP) mapped node features to 64-dimensional embeddings (hidden size 128, dropout 0.5; GAT with four heads in the first layer), and pairs were scored with a dot-product decoder. Models were trained with binary cross-entropy and Adam (learning rate 10^{-3} , at most 60 epochs, early stopping on validation ROC-AUC with patience 12) in PyTorch Geometric. Seven feature sets combined Biomni descriptors, ESM-2 embeddings and Node2Vec embeddings. Node2Vec [22] (128 dimensions, $p = q = 1$, 80 walks of length 10 per node, window 5) was refitted for every split on training edges only. Non-learned baselines were common neighbours, Adamic-Adar, resource

allocation and preferential attachment [26, 27], the degree of the partner protein in the training graph, and the cosine similarity of ESM-2 embeddings.

2.6 Evaluation protocols

Each protocol used 10 random splits (seeds 0–9) and the same splits for every model and feature set (Fig. 1).

P1, edge split. Edges were split 80/10/10 into training, validation and test sets with an equal number of uniformly sampled non-edges as negatives; both proteins of a test pair generally remain in the training graph.

P2, held-out proteins with uniform negatives. 10% of connected proteins were held out for validation and 10% for testing together with all their edges, so that at evaluation time they have no edges in the message-passing graph, exactly like interaction-orphan proteins. Test positives are edges between a test protein and a training protein (class C2 of Park and Marcotte [15]); pairs of two test proteins (C3) are reported separately. Negatives were uniform random non-edges with the same node-type constraint.

P3, held-out proteins with degree-matched negatives. As P2, but each endpoint of a negative pair was sampled in proportion to its frequency among positive pairs, so that degree alone cannot separate positives from negatives [17].

We report ROC-AUC and average precision (mean $\pm$ s.d. over splits). Each configuration was compared with the best one by paired *t*-tests and Wilcoxon signed-rank tests over splits, with Benjamini–Hochberg correction across the family of comparisons.

2.7 External validation on withheld proteins

Proteins that were never submitted to STRING but are mappable to STRING v12 provide associations that no model saw during development. For each such protein that is isolated in the development network, we scored all candidate partners (all connected proteins and the other isolated proteins, excluding identical sequences and pairs mapped to the same STRING protein) with the models trained on the development network and compared the rankings with its STRING v12 links (combined score ≥ 400). Metrics were per-protein ROC-AUC, precision among the five top-ranked partners (P@5), whether a true partner appears among the top ten, and pooled ROC-AUC and precision across all pairs.

2.8 Candidate scoring and re-ranking

Final scores were averaged over the ten models trained on the P3 splits, using the configuration selected on the development network (Biomni+ESM-2, MLP encoder; it is within the statistically tied group on the final network). Because dot-product scores favour a few partners with large embedding norms, five ranking variants were declared before evaluation: the raw score; cosine similarity of model embeddings; a hubness-corrected cosine (CSLS, $k = 10$) [28]; ESM-2 cosine similarity; and rank blends of the model cosine with ESM-2 cosine. The variant with the highest mean per-protein P@5 on the internal P3 test proteins was pre-selected; all variants were then evaluated once on the external set.

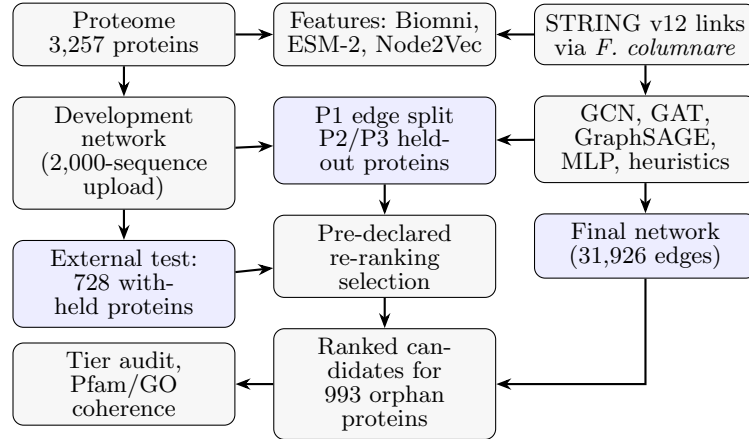


Fig. 1 Study design. Models and ranking choices were developed on the network built from the original 2,000-sequence STRING upload; proteins withheld by the upload limit served as an external test set; the final network built from all proteins was used for the reported benchmark and the released candidate list.

2.9 Biological evaluation

Six sets of Pfam families associated with surface exposure or secretion were fixed in advance by accession (T9SS C-terminal sorting domains, outer-membrane proteins, TonB-dependent/SusCD uptake systems, T9SS and gliding machinery, adhesin and surface-repeat domains, type VI secretion; Supplementary Table A3). Enrichment of these families by Biomni tier was tested with Fisher’s exact test (High versus other tiers) and Spearman correlation with tier rank. Functional coherence of predicted pairs was measured as the fraction of pairs (both proteins annotated) sharing a Pfam family, a Pfam clan or a GO term, compared with 50,000 random candidate pairs and with all edges of the final network. Tier-pair composition of predictions was compared with that of the recovered STRING links of the external proteins.

2.10 Software and reproducibility

Analyses used Python 3.12.9, PyTorch 2.9.1, PyTorch Geometric 2.7.0, scikit-learn 1.6.1, gensim 4.4.0, fair-esm 2.0.0 and pyhmmer 0.12.3. All code, configuration files, per-split indices, results tables and the candidate lists are released (see Data availability).

3 Results

3.1 On random edge splits, neighbourhood heuristics match or beat GNNs

Under the edge-split protocol P1, every GNN configuration reached ROC-AUC of at least 0.874 (Fig. 2a), and Biomni descriptors alone reached 0.895 ± 0.004 . Yet the best parameter-free heuristic (resource allocation, 0.969 ± 0.002) matched or exceeded the

Table 1 Data used in this study.

	Development network	Final network
Proteins (nodes)	3,257	3,257
Proteins mapped to STRING v12	1,760	2,533
Edges (combined score ≥ 400)	16,315	31,926
Proteins without edges	1,650	993
not submitted to STRING	1,257	0
no STRING match	240	724
lower-scoring paralogue of a mapped protein	153	235
mapped, but no link with score ≥ 400	0	34

best GNN configuration (Node2Vec with GCN, 0.956 ± 0.005). Because both proteins of nearly every test pair keep other edges in the training graph, this protocol measures how well a model reproduces local network structure, which simple heuristics already capture.

3.2 Holding out proteins removes all topological signal and exposes degree bias

When whole proteins were held out (P2), the topological methods failed as expected: neighbourhood heuristics scored exactly 0.5 because held-out proteins have no neighbours, and Node2Vec features were at or below chance with GCN and GAT (Fig. 2b). With uniformly sampled negatives, however, ranking pairs by the degree of the training-graph partner reached 0.860 ± 0.004 ROC-AUC and outperformed every trained model (best Biomni+ESM-2 with MLP, 0.835 ± 0.025). Uniform negatives therefore mostly reward predicting that new proteins attach to hubs.

With degree-matched negatives (P3), partner degree fell to 0.538 ± 0.007 and Node2Vec to chance (Fig. 2c, Table 2). The remaining signal came from sequence embeddings: the four feature sets containing ESM-2 reached 0.767–0.777 with the graph-free MLP (best Biomni+ESM-2+Node2Vec with MLP, 0.777 ± 0.025 ; ESM-2 alone 0.767 ± 0.020), and their differences were not significant after correction (BH-adjusted Wilcoxon $p \geq 0.07$). Node2Vec alone stayed at chance (at most 0.524), Biomni descriptors alone reached 0.672 ± 0.026 , and adding Biomni descriptors to ESM-2 changed ROC-AUC by +0.003 (paired Wilcoxon $p = 0.037$, uncorrected). For every feature set containing ESM-2, the graph-free MLP was at least as good as any GNN encoder. ESM-2 650M was marginally better than the 35M model (Supplementary Table A1).

3.3 External validation confirms the held-out-protein estimate

728 proteins that were isolated in the development network only because they were never submitted to STRING have 22,500 STRING v12 links to candidate partners (base rate 0.012). Scored by models trained before these links were known, they gave a pooled ROC-AUC of 0.708 (Table 3, Fig. 3), in line with the internal P3 estimate for the same model and network (0.715), and well above partner degree (0.558). Biomni

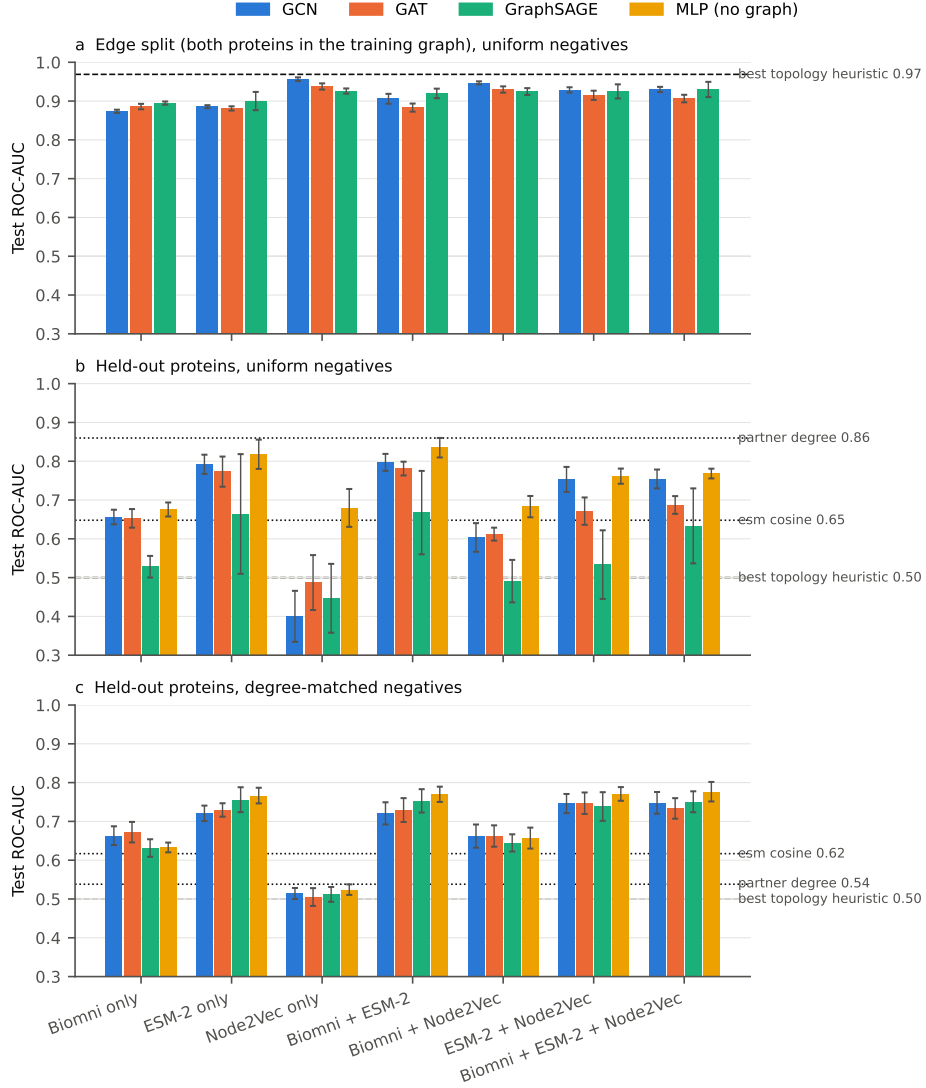


Fig. 2 Benchmark on the final network (10 splits; bars, mean test ROC-AUC; whiskers, s.d.). (a) P1, random edge split with uniform negatives (GNN encoders only). (b) P2, held-out proteins with uniform negatives (test pairs between a held-out and a training protein). (c) P3, held-out proteins with degree-matched negatives. Horizontal lines mark non-learned baselines.

descriptors did not improve on ESM-2 alone in this set (per-protein ROC-AUC 0.653 versus 0.650; Wilcoxon $p = 0.99$).

Per-protein partner lists were dominated by a few partners with large embedding norms (raw-score P@5 0.089). Among the pre-declared re-ranking variants, ESM-2 nearest neighbours had the highest internal P@5 and were therefore pre-selected; externally they again gave the best partner lists (P@5 0.135, a true partner in the top ten

Table 2 Held-out proteins with degree-matched negatives (P3), final network. Test ROC-AUC, mean $\pm$ s.d. over 10 splits; best value per row in bold.

Feature set	GCN	GAT	GraphSAGE	MLP
Biomni	0.663 $\pm$ 0.024	0.672 $\pm$ 0.026	0.631 $\pm$ 0.023	0.633 $\pm$ 0.013
ESM-2	0.721 $\pm$ 0.020	0.729 $\pm$ 0.017	0.756 $\pm$ 0.032	0.767 $\pm$ 0.020
Node2Vec	0.514 $\pm$ 0.014	0.505 $\pm$ 0.023	0.512 $\pm$ 0.019	0.524 $\pm$ 0.014
Biomni+ESM-2	0.721 $\pm$ 0.029	0.729 $\pm$ 0.031	0.753 $\pm$ 0.030	0.770 $\pm$ 0.020
Biomni+Node2Vec	0.662 $\pm$ 0.030	0.662 $\pm$ 0.028	0.645 $\pm$ 0.022	0.657 $\pm$ 0.027
ESM-2+Node2Vec	0.746 $\pm$ 0.025	0.747 $\pm$ 0.028	0.738 $\pm$ 0.037	0.771 $\pm$ 0.018
Biomni+ESM-2+Node2Vec	0.748 $\pm$ 0.028	0.733 $\pm$ 0.026	0.751 $\pm$ 0.027	0.777 $\pm$ 0.025
Best topology heuristic		0.500 $\pm$ 0.000		
Partner degree		0.538 $\pm$ 0.007		
ESM-2 cosine similarity		0.617 $\pm$ 0.007		

Table 3 External validation on 728 proteins withheld from the development network (22,500 STRING v12 links). Models were trained on the development network (Biomni+ESM-2 unless stated).

Ranking	Per protein			Pooled	
	AUC	P@5	Hit@10	AUC	P@0.1%
Model, Biomni+ESM-2	0.653	0.089	0.216	0.708	0.195
Model, ESM-2 only	0.650	0.091	0.221	0.721	0.356
ESM-2 cosine (pre-selected)	0.638	0.135	0.404	0.595	0.059
Blend (0.5)	0.670	0.118	0.353	0.733	0.199
Partner degree	0.547	0.090	0.317	0.558	0.105
Random	0.500	0.012	–	0.500	0.012

for 40% of proteins), but the worst pooled ranking (ROC-AUC 0.595). Blending model cosine with ESM-2 cosine gave the best per-protein ROC-AUC (0.670) with intermediate P@5 (0.118) and pooled ROC-AUC 0.733. The internal ordering of variants predicted the external ordering, supporting the use of P3 for model selection.

3.4 Biomni priority tiers do not single out surface or secreted protein families

Of the 154 proteins carrying a pre-specified surface or secretion family, the highest fraction fell in the Medium tier (12.4%) rather than the High tier (3.0%; Fig. 4a). High-tier proteins were not enriched for these families (odds ratio 0.61, BH-adjusted $p = 0.85$); the weak positive trend over tiers (Spearman $\rho = 0.11$) was carried by the Medium tier, and no TonB-dependent or SusCD protein was ranked High.

In the model’s top predictions, pairs of Medium- and High-tier proteins were over-represented relative to the candidate pool. The external set shows that this is not a property of the true associations: compared with the recovered STRING links of the same proteins, the model’s top-five partners over-represented Medium–Medium pairs 3.0-fold and High–High pairs 2.4-fold, and under-represented Very Low–Very Low pairs (0.59 $\times$; Fig. 4b). Tier homophily in predicted links should therefore not be read as biological evidence.

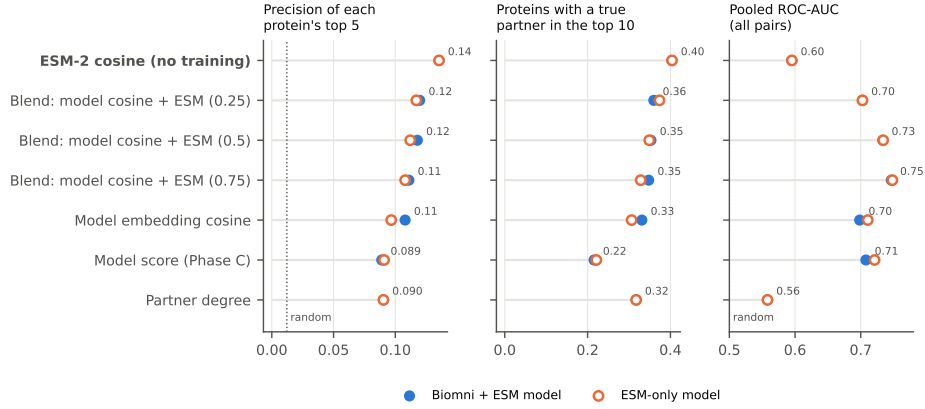


Fig. 3 External validation of ranking variants. Filled markers, Biomni+ESM-2 model; open markers, ESM-2-only model. The bold variant was selected in advance on the internal P3 splits. Dotted lines, random expectation.

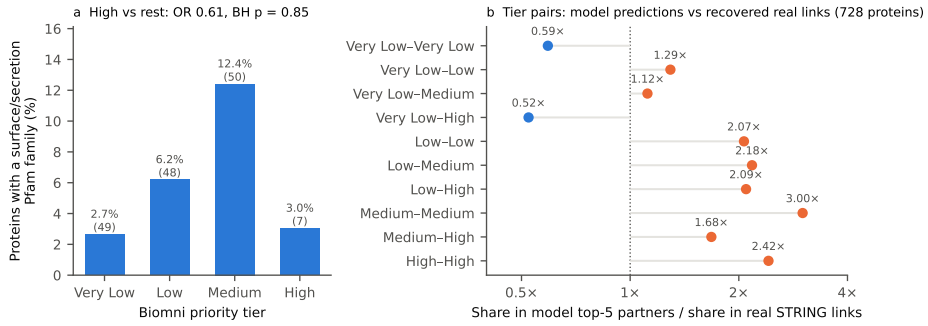


Fig. 4 Audit of Biomni priority tiers. (a) Fraction of proteins carrying a pre-specified surface or secretion Pfam family, by tier (counts in parentheses). (b) Tier-pair composition of the model's top-five partners relative to the recovered STRING v12 links of the same 728 external proteins.

3.5 Predicted partners are functionally coherent

On the final network, the top 2,000 predicted pairs (at most five per protein) shared a GO term (16.9% for isolated–connected and 57.7% for isolated–isolated pairs) and a Pfam clan (7.5% and 26.4%) far more often than random candidate pairs (5.6% and 24.8% for GO terms; all BH-adjusted $p < 0.001$), at rates comparable to the edges of the network itself (19.8% GO, 6.7% clan), and higher for isolated–isolated pairs (Fig. 5). Because ESM-2 embeddings encode homology, part of this coherence reflects sequence similarity, and it supports plausibility rather than independent confirmation. The released candidate list covers the 993 proteins without edges in the final network, annotated with Pfam domains, GO terms, Biomni descriptors and identical-sequence groups.

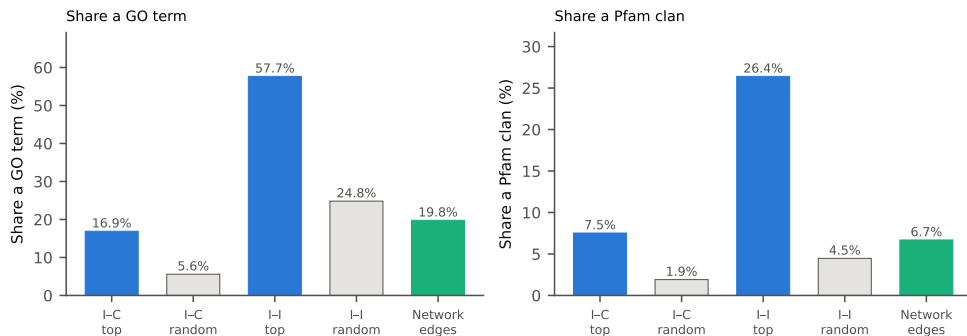


Fig. 5 Functional coherence of the top 2,000 predicted pairs on the final network (at most five per protein) compared with random candidate pairs and with the network’s own edges. I-C, isolated-connected pairs; I-I, isolated-isolated pairs.

4 Discussion

The ranking of methods in this study was determined by the evaluation protocol rather than by model architecture. On random edge splits, GNNs were matched or outperformed by neighbourhood-counting heuristics; with whole proteins held out but uniform negatives, a degree heuristic beat every model; and only with degree-matched negatives did a stable picture emerge, in which sequence embeddings carried the signal and graph structure carried none. These observations agree with general analyses of link-prediction and pair-input evaluation [15–17] and with the finding that PLM embeddings account for most of the attainable performance in leakage-controlled interaction prediction [19]. For reverse vaccinology, where the proteins of interest are often exactly those without recorded interactions, the practical implication is that network-topology methods add nothing for such proteins and that reported performance must come from a held-out-protein protocol with degree-controlled negatives.

The external validation is the most direct test of this protocol. Because the upload limit withheld the associations of 1,257 proteins from the development network, models could be scored on associations that no step of development had seen, and the internal estimate (0.715) matched the external result (0.708). The same set showed that the best tool for a per-protein partner list was the simplest: nearest neighbours in ESM-2 space. Trained models remain useful for ranking pairs across proteins, which matters when deciding which predictions to follow up, and blending the two gives a balanced ranking. We report the pre-declared choice together with the alternatives so that users can select the ranking suited to their purpose.

Biomni descriptors added at most a marginal amount of information to ESM-2 internally and none externally, and the priority tiers derived from them did not identify the outer-membrane, TonB-dependent and T9SS cargo families that dominate the *Flavobacterium* virulence literature [7, 9]. Agent-derived priority scores are convenient, but they encode heuristic weightings whose behaviour should be audited against independent annotation before they are used to select antigens. Releasing the agent’s version, prompts and generated code is necessary for such outputs to be reproducible.

Several limitations apply. Associations were transferred from a related strain, so they inherit both the coverage of *F. columnare* in STRING and the uncertainty of homology transfer; STRING associations are functional, not necessarily physical. The external set consists of proteins withheld for a technical reason, and they are shorter on average than the submitted ones; their associations include conserved complexes such as the ribosome, which are easier to predict than those of typical antigen candidates. ESM-2 similarity partly encodes homology, so functional coherence of predictions is evidence of plausibility rather than of novel biology. Finally, the analysis covers one proteome and makes no claim about transfer to other pathogens, and no prediction has been tested experimentally; the candidate list is a set of ranked hypotheses for follow-up.

5 Conclusions

For proteins without known interactions in a non-model fish pathogen, protein language model embeddings provide the usable signal for association prediction, graph topology provides none, and agent-derived vaccine priorities do not track independent surface and secretion annotations. A held-out-protein protocol with degree-matched negatives gave performance estimates that held on externally withheld proteins. We release the benchmark, code and an annotated candidate list as a reproducible basis for reverse vaccinology in *Flavobacterium*.

Supplementary information. Supplementary Tables A1–A4 are provided in the Appendix. Supplementary File S1 documents the Biomni run [AUTHOR INPUT: to be supplied].

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Declarations

- **Funding:** [AUTHOR INPUT: funding sources and grant numbers].
- **Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable; computational study of publicly available or author-generated bacterial sequence data.
- **Data availability:** The proteome, both networks, mappings, Pfam/GO annotations, ESM-2 embeddings, per-split indices, all result tables and the candidate lists are available at https://github.com/CEL-lab/USDA_Protein [AUTHOR INPUT: archived release DOI (e.g. Zenodo) and genome accession]. STRING v12 data are available from <https://string-db.org>.
- **Code availability:** All analysis scripts are in the Code/ directory of the repository above.
- **Author contributions:** [AUTHOR INPUT: CRediT contributions].

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Appendix A Supplementary tables

Table A1 ESM-2 model size, development network, P3 (held-out proteins, degree-matched negatives). Test ROC-AUC, mean $\pm$ s.d. over 10 splits.

Feature set	Encoder	ESM-2 35M	ESM-2 650M
ESM-2	GraphSAGE	0.710 ± 0.026	0.716 ± 0.027
ESM-2	MLP	0.727 ± 0.023	0.724 ± 0.023
Biomni+ESM-2	GraphSAGE	0.709 ± 0.018	0.718 ± 0.031
Biomni+ESM-2	MLP	0.733 ± 0.020	0.742 ± 0.017
ESM-2+Node2Vec	GraphSAGE	0.705 ± 0.037	0.730 ± 0.027
ESM-2+Node2Vec	MLP	0.720 ± 0.026	0.729 ± 0.023
Biomni+ESM-2+Node2Vec	GraphSAGE	0.710 ± 0.022	0.714 ± 0.044
Biomni+ESM-2+Node2Vec	MLP	0.721 ± 0.029	0.725 ± 0.021
ESM-2 cosine similarity	–	0.610 ± 0.008	0.635 ± 0.006

Table A2 Pre-declared re-ranking variants: internal P3 test proteins (final network) and external validation (development network).

Variant	Internal (final network)			External		
	P@5	Prot. AUC	Pooled AUC	P@5	Prot. AUC	Pooled AUC
ESM-2 cosine	0.125	0.646	0.601	0.135	0.638	0.595
Blend cosine/ESM-2 (0.25)	0.116	0.663	0.682	0.120	0.659	0.703
Blend CSLS/ESM-2 (0.5)	0.110	0.674	0.711	0.116	0.671	0.734
Blend cosine/ESM-2 (0.5)	0.109	0.673	0.711	0.118	0.670	0.733
Blend cosine/ESM-2 (0.75)	0.099	0.671	0.725	0.111	0.668	0.746
Model cosine, CSLS	0.086	0.660	0.693	0.107	0.661	0.701
Model cosine	0.084	0.659	0.690	0.108	0.660	0.698
Model score	0.079	0.654	0.712	0.089	0.653	0.708

Table A3 Pfam family sets fixed in advance for the biological evaluation.

Family set	Pfam families
T9SS cargo (CTD)	Por_Secre_tail (PF18962), CHU_C (PF13585)
Outer membrane protein	OmpA (PF00691), OmpA_like (PF16961), OMP_b-brl_3 (PF14905), OMP_b-brl_2 (PF13568), OMP_b-brl (PF13505), OmpH (PF03938), OEP (PF02321), Omp85 (PF01103), Omp85_2 (PF19143), POTRA (PF07244), Omp28 (PF11551), Toluene_X (PF03349), LptD_2 (PF19838), LptD_N (PF03968), Gly-zipper_Omp (PF13488), YfiO (PF13525)
TonB / SusCD uptake	Plug (PF07715), TonB_dep_Rec_b-barrel (PF00593), SusD_RagB (PF07980), SusD-like (PF12741), SusD-like_2 (PF12771), SusD-like_3 (PF14322), TonB_C (PF03544)
T9SS / gliding machinery	PorP_SprF (PF11751), PorV (PF19572), T9SS_plug_1st (PF17116), SprA_N (PF14349), SprB (PF13573), GldM_3rd (PF21602), GldM_2nd (PF21601), GldM_4th (PF12080), GldM_1st (PF12081), GldL_N (PF22827), GldN (PF19841), GldB_lipo (PF25594), GldD_lipo (PF25593), GldH_lipo (PF14109), GldC-like (PF19937)
Adhesin / surface repeat	DUF2807 (PF10988), Cleaved_Adhesin (PF07675), Big_5 (PF13205), PKD (PF00801), PKD_4 (PF18911), fn3 (PF00041), Laminin_G_3 (PF13385), RHS_repeat (PF05593)
T6SS (type VI secretion)	TssD (PF17642), TssN (PF17555), TssO (PF17561), TssC (PF17541)

Table A4 Surface and secretion family sets by Biomni priority tier (fraction of proteins in each tier; Fisher test High versus other tiers, BH-adjusted).

Family set	<i>n</i>	Very Low	Low	Medium	High	OR High (BH <i>p</i>)
T9SS cargo (CTD)	43	0.6%	2.2%	3.5%	0.4%	0.31 (0.85)
Outer membrane protein	44	0.8%	0.9%	4.7%	1.7%	1.32 (0.96)
TonB / SusCD uptake	31	0.5%	1.9%	1.7%	0.0%	0 (0.85)
T9SS / gliding machinery	21	0.2%	0.6%	2.7%	0.4%	0.66 (1.00)
Adhesin / surface repeat	14	0.2%	0.9%	0.7%	0.4%	1.01 (1.00)
T6SS (type VI secretion)	13	0.6%	0.3%	0.0%	0.0%	0 (1.00)
Any surface/secretion family	154	2.7%	6.2%	12.4%	3.0%	0.61 (0.85)